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Task Description:

If we fantasize for a moment, we can envision a new class of materials based on Nanocrystal superlattice (NCLS). These materials would be analogous to their atomic and molecular crystal cousins, but the lattice spacings roughly one order of magnitude bigger. The key factor in achieving this fantasy is the creation of the superlattice (SL) crystal. We have argued that digestive ripening will provide monodisperse (NC) of almost any material and these can then form SL crystals. This has been achieved in a few simple cases, but the SL crystals are by no measure macroscopic. Here again, we propose more careful, yet standard, crystal growth procedures, and expect some success. However, it is now well known, as a result of past microgravity research, that crystals can grow better in the microgravity environment. Therefore, we would eventually like to grow SL crystals in space.
In a microgravity environment gravity induced flows are greatly suppressed. These flows can cause gradients in temperature and composition in the solution which in turn cause nonuniformities, defects, and solvent inclusions in the growing crystals. Great care must be taken in Earth-based crystal growth situations such as milli-kelvin temperature control and careful stirring to avoid these problems. This is especially true for growth of polymeric or protein crystals, which have been grown much more successfully in microgravity. Since these problems increase with the size of the molecule, it is reasonable to expect NCSLs with their large 'molecules' might be equally difficult to grow on Earth, but equally remedied by microgravity, as polymeric and protein crystals.

To prepare large NCSL materials and assess their physical (including optical, melting points, lattice energies) and chemical properties.

These new materials may lead to new electronic or optical devices, and may be important for future computers and information storage.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Crystals in Nature are built from atoms or molecules arranged in a specific order. If one has a colloidal solution containing particles with one and the same size, they can behave as 'atoms' and construct their own crystal lattice (NCSL). The structure and properties of this material have nothing to do with the characteristics of the colloidal particles by themselves, but rather they are dependent upon the mutual interactions between the 'nano-atoms' forming the crystal. Many properties of the material are strongly contingent on the sizes of the constituting particles, and therefore, its characteristics can be tuned by changing particles' sizes. The key step in realizing this idea is the ability to synthesize colloids with a very narrow size distribution.
The research has been focused on the improvement and study of the already reported conditions for the digestive ripening process—a remarkable procedure for tailoring nanoparticles' sizes recently discovered by our group. Effort has been spent in synthesizing large amounts of monodispersed gold colloids and for studying the effects of various ligands for digestive ripening in order to reach the goal of synthesizing NCSL materials. We have made significant progress in understanding the important steps of the digestive ripening process as described below.

Two experimental methods have been used for the preparation of the colloids: the inverse micelles technique and the solvated metal atom dispersion (SMAD) method. All experiments have been conducted multiple times and reproduction of the data has been obtained. Briefly the procedures and results from both techniques will be described below.

Inverse micelles method

Inverse micelles are formed in systems containing both hydrophobic and hydrophilic solvents. The surfactant forms structures in which its hydrophilic heads surround the hydrophilic solvent, while the hydrophobic parts point towards the hydrophobic solvent. In fact the micelles act as microreactors in which the process of particles formation and growth take place. Since micelles are expected to have almost uniform size, the resulted particles should roughly have uniform size as well. This procedure is widely employed in practice.

In our case the surfactant is didodecyldimethylammonium bromide (DDAB). Toluene and water are used as hydrophobic and hydrophilic solvents, respectively. Gold chloride ($AuCl_3$) is dissolved in the DDAB/toluene mixture followed by a subsequent reduction with an aqueous solution of sodium borohydride ($NaBH_4$). In this way the as-prepared Au-colloids are synthesized. All synthetic steps are conducted in an inert box to prevent influence of moisture and air. The most frequently utilized techniques by us for the characterization of the synthesized colloids are transmission electron microscopy (TEM) and ultraviolet spectroscopy (UV-VIS).

The as-prepared colloids have sizes of the particles varying from 20 to 150 nm. The particles have various regular shapes (rod-like, hexagonal, spherical). Previously, we have reported that using dodecanethiol as a capping and digestive ripening agent results in monodisperse gold nanoparticles which are ordered in both two dimensional and three dimensional arrays. We have now extended this study and used octane, decane. And hexadecane—thiols also as digestive ripening agents. We observed that all of them are good digestive agents leading to highly monodisperse particles. However, their solubility and the ability to form 2D or 3D lattices at room temperature is greatly affected by the alkyl chain length of the thiol used. For example, using using octanethiol forms only 3D SLs at room temperature. Colloids digested with decane and dodecanethiol, on the other hand, show the coexistence of both separate particles and 3D lattices, while hexadecanethiol digestion leads to the formation of a stable colloid with separate particles. The UV-Vis spectral features at room temperature for the above colloids are: (1) octanethiol and decanethiol digested particles showing a large tail above 700 nm apart from the gold plasmon band at 520 nm, (2) Dodecanethiol digested particles reveal a shoulder at 630 nm apart from the plasmon band and (3) hexadecane thiol particles show only the gold plasmon band. The additional features in the UV-vis spectra (large tail in octane and decane thiol case and the shoulder in dodecanethiol case) are attributed to the 3D SLs present in these colloids. However, at higher temperatures all the above samples show only the gold plasmon band indicating that the 3-D lattices are broken apart and only the gold-thiol monomers are present. Another interesting point is that the thiol capped gold particles are readily soluble in toluene while they are insoluble in many alcohols. These features are similar to the solubility behavior of normal molecular materials and have been used to crystallize many of the molecular materials by carefully controlling the temperature of the solvent and by using a combination of solvents. Thus the similarity of the solubility of thiol capped gold particles to normal molecular materials is quite promising and might provide us many clues to crystallize them into SLs.

Our studies with ligands other than thiols for digestive ripening offered us vital insights into the mechanistic aspects of digestive ripening. We have identified three important steps in digestive ripening, and they are (1) breaking the

polydisperse colloid into smaller size particles upon addition of the ligand, (2) isolating this colloid from the reaction side products, and finally (3) heating this isolated colloid in the presence of the ligand to form a nearly monodisperse colloid. We found that apart from thiols, amines, silanes and phosphines were similarly efficient digestive ripening agents and the sizes of the nearly monodisperse particles depended on the ligand used for digestive ripening. The ligand-gold particle size dependency could be rationalized semi-qualitatively by invoking the hard and soft acid and base theory which suggests that a soft acid like gold likes to interact with softer bases like RSH and R3P than a harder acid like RNH2. This enabled us to control the individual sizes of the gold nanoparticles thus paving the way for preparing superlattice crystals of different size particles.

Solvated metal atom dispersion (SMAD) method

The SMAD technique is a well studied method for the synthesis of a large variety of air and moisture sensitive metals and organometallic compounds. It has been widely used and investigated in our laboratory for many years. It has the great advantage of ability to produce large quantities of the desired material.

The SMAD method is based on the vaporization of metal under high vacuum and simultaneous condensation of the vaporized atoms with the vapors of solvent on the walls of a reactor at liquid nitrogen temperature (77K). Warming of the matrix and formation of the final colloid follow this. Several factors play a crucial role for the characteristics of the particles produced. These include polarity of the solvent, rate of vaporization, amount of solvent, matrix warm-up rate.

Gold colloids in acetone are well studied in our laboratory. That is why the as-prepared colloid is synthesized in an acetone matrix by vaporization of gold metal and codeposition with the acetone vapors in liquid nitrogen temperature. In order to prevent aggregation of the particles, the colloid has been warmed up in the presence of dodecanethiol present as a toluene solution. At this stage, the colloid is characterized with non-uniform, undefined shaped particles with sizes ranging from 10 to 30 nm. The acetone is then removed under vacuum, and the resulting colloid is subsequently digested under reflux and argon atmosphere. A drastic change of shape and sizes of particles is observed. They acquire sherical shape and sizes are reduced to the ‘magic’ number of ~4.5 nm. These particles tend to form SLs which sometimes are as big as 100 μm.

Although the sizes of the individual particles are similar, the packing nature of the inverse micelle prepared and SMAD prepared colloids are quite different. While the inverse micelle preparation leads to the arrangement of particles in an fcc type lattice with ABCABC…. type ordering, SMAD colloids result in hcp lattice with ABAB…ordering. Currently efforts are under way in ascertaining the facts governing the packing nature of these colloids. We are also investigating the effects of different substrates on the packing nature of these colloids. The above studies will have important bearing in understanding the interactions between these particles and consequently provide the clues to prepare isolatable superlattice crystals.

We have recently been able to extend the SMAD procedure to producing large quantities of monodisperse nanoparticles to a host of new and exciting materials. These materials include silver and copper metals; and ZnS, ZnSe, CdS, CdTe, PbS, PbSe, SnS semiconductors. These materials have expressed excellent super lattice formation, which can be modified drastically by the ligand-stabilizing agent used. For example, using dodecane thiol we have grown three-dimensional triangular superlattices of silver nanoparticles, while using phosphorus based ligands produced circular SLs, both of micron sizes. These new materials extend the usefulness of this procedure as interesting new characteristics and phenomena can be observed for each elemental combination much like the vast richness of nature is composed of the contributions from expanse of the periodic table. A few examples of these properties include the tunable light absorption qualities of these materials that depend heavily on the control of synthesis of the materials and their crystallization. Also significant fluorescence emission has been observed by the semiconductor particles, which is also highly dependent on the quality of the product. The large quantities available through the SMAD method is crucial to being able to study the physics and chemistry of these new materials and to combinations of them as we are preparing to do in our laboratory.

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